Khaled Younes

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Education

Stanford University — Stanford, CA
PhD Mechanical Engineering — 2026

University of Waterloo — Waterloo, ON
MASc Mechanical Engineering — 2021
BASc Mechanical Engineering — 2017

Research Appointments

Canada Impact+ Postdoctoral Scholar
University of Waterloo — 2026 – Present

Research Assistant
University of Waterloo — 2017 – 2019

Awards and Scholarships

Canada Impact+ Research Training Award (CIRTA) – Postdoctoral	2026
Early Career Award (Ultrafast Imaging and Tracking Instrumentation, Methods and Applications Conference)	2023
Finalist for the Alumni Gold Medal	2021
Finalist for the Governor General's Gold Medal	2021
Postgraduate Scholarship – Doctoral (NSERC PGS D)	2021
Canada Graduate Scholarship – Master's (NSERC CGS M)	2020
University of Waterloo President's Graduate Scholarship	2020
Sandford Fleming Foundation Teaching Assistantship Excellence Award	2020
Engineering Excellence Master's Fellowship	2019
Engineering Dean's Entrance Award	2019
John Deere Limited Scholarship	2017
First In Class Engineering Scholarship	2016
University of Waterloo President's Scholarship	2013

Teaching Experience

Graduate Teaching Assistant at Stanford
Engineering Thermodynamics (ME 30) — Winter 2024, Fall 2024, Winter 2026
Gas Turbine Design Analysis (ME 357) — Fall 2025

Graduate Teaching Assistant at Waterloo
Fluid Mechanics 1 (ME 351) — Spring 2020
Heat Transfer (ME 353) — Winter 2020

Residence Tutor at Claudette Millar Hall
Calculus 2 for Engineering (Math 118) Winter 2018
Calculus 1 for Engineering (Math 116), Linear Algebra for Engineering (Math 115) Fall 2017

General Teaching Assistant at Waterloo Engineering Endowment Foundation
Mechanics (Phys 115), Calculus 2 for Engineering (Math 118) Winter 2014

Refereed Journal Articles

Published

J13. J. Fan, T. Yoon, G. Vignat, H. Li, **K. Younes**, A. Majumdar, P. Muhunthan, D. Sokaras, T. Weiss, I. Rajkovic, M. Ihme, "Supercritical Ethanol-CO_2 Mixtures Exhibit Microscopic Immiscibility: A Combined Study Using X-ray Scattering and Molecular Dynamics Simulations," *The Journal of Physical Chemistry Letters*, vol. 16, no. 27, 2025. doi:10.1021/acs.jpclett.5c01293

J12. P. Muhunthan, A. Majumdar, **K. Younes**, G. Vignat, H. Li, I. Rajkovic, T. Weiss, D. Sokaras, M. Ihme, "A self-consistent analysis of cluster morphology in supercritical carbon dioxide from Small Angle X-ray Scattering," *Chemical Physics Letters*, vol. 876, 142190, 2025. doi:10.1016/j.cplett.2025.142190

J11. P. Muhunthan, H. Li, G. Vignat, E. R. Toro, **K. Younes**, Y. Sun, D. Sokaras, T. Weiss, I. Rajkovic, T. Osaka, I. Inoue, S. Song, T. Sato, D. Zhu, J. L. Fulton, M. Ihme, "A versatile pressure-cell design for studying ultrafast molecular-dynamics in supercritical fluids using coherent multi-pulse x-ray scattering," *Review of Scientific Instruments*, vol. 95, 013901, 2024. doi:10.1063/5.0158497

J10. **K. Younes**, M. Poli, P. Muhunthan, S. Ermon, M. Ihme, "Autonomous screening of complex phase spaces using Bayesian optimization for SAXS measurements," *Nuclear Instruments and Methods in Physics Research Section A*, vol. 1057, 168719, 2023. doi:10.1016/j.nima.2023.168719

J9. T. Zirwes, G. Vignat, E. R. Toro, E. Boigné, **K. Younes**, D. Trimis, M. Ihme, "Improving volume-averaged simulations of matrix-stabilized combustion through direct X-ray μCT characterization: Application to NH_3/H_2-air combustion," *Combustion and Flame*, vol. 257, part 2, 113020, 2023. doi:10.1016/j.combustflame.2023.113020

J8. G. Vignat, T. Zirwes, E. R. Toro, **K. Younes**, E. Boigné, P. Muhunthan, L. Simitz, D. Trimis, M. Ihme, "Experimental and numerical investigation of flame stabilization and pollutant formation in matrix stabilized ammonia-hydrogen combustion," *Combustion and Flame*, vol. 250, 112642, 2023. doi:10.1016/j.combustflame.2023.112642

J7. **K. Younes** and J.-P. Hickey, "Mean velocity scaling of high-speed turbulent flows under nonadiabatic wall conditions," *AIAA Journal*, vol. 61, no. 4, 2023. doi:10.2514/1.J062547

J6. **K. Younes**, B. Gibeau, S. Ghaemi, J.-P. Hickey, "A Fuzzy Cluster Method for Turbulent/Non-Turbulent Interface Detection," *Experiments in Fluids*, vol. 62, no. 73, 2021. doi:10.1007/s00348-021-03169-9

J5. J.-P. Hickey, **K. Younes**, M. Yao, D. Fan, J. Mouallem, "Targeted turbulent structure control in wall-bounded flows via localized heating," *Physics of Fluids*, vol. 32, 035104, 2020. doi:10.1063/1.5144387 **(Featured Article; Highlighted in Scilight)**

J4. **K. Younes** and J.-P. Hickey, "Fluidic Thrust Shock-Vectoring Control: A Sensitivity Analysis," *AIAA Journal*, vol. 58, no. 4, 2020. doi:10.2514/1.J058922

J3. **K. Younes** and J.-P. Hickey, "Effects of shear layer growth on the indirect noise in compound nozzles," *Journal of Sound and Vibration*, vol. 468, 115090, 2020. doi:10.1016/j.jsv.2019.115090

J2. **K. Younes** and J.-P. Hickey, "Indirect noise prediction in compound, multi-stream nozzle flows," *Journal of Sound and Vibration*, vol. 442, pp. 609–623, 2019. doi:10.1016/j.jsv.2018.10.061

J1. J.-P. Hickey and **K. Younes**, "Path to turbulence in a transitional asymmetric planar wake," *Physics of Fluids*, vol. 31, 104107, 2019. doi:10.1063/1.5118891

Accepted

A3. M. Asplund, H. Liu, T. Yoon, Y. Liu, R. Wallick, S. Bhattacharyya, X. Cheng, S. Eisenberg, R. England, M. Graßl, ..., **K. Younes**, B. Fingerhut, M. Ihme, T. Elsaesser, M.-F. Lin, B. Schwartz, J. Rouxel, L. Young, "Direct measurement of the structure of the hydrated electron," *Nature Communications*, 2026.

A2. A. Majumdar, H. Li, Y. Sun, S. Song, G. Vignat, P. Muhunthan, T. Sato, **K. Younes**, A. Halavanau, A. Lutman, D. Sokaras, D. Zhu, M. Ihme, "Ultrafast collective dynamics emerges from nanoscale structural heterogeneities in supercritical fluids," *Nature Communications*, 2026.

A1. T. Yoon, H. Li, Y. Sun, A. Majumdar, P. Muhunthan, G. Vignat, S. Song, T. Sato, S. She, J. Fan, Y. Li, **K. Younes**, A. Halavanau, A. Lutman, A. Maurya, D. Zhu, M. Ihme, "Revealing Ultrafast Multiscale Dynamics of Liquid CO_2 Radiolysis with Time-Resolved X-ray Pump/X-ray Probe Scattering using Split-and-Delay Optics," *Structural Dynamics*, 2026.

Manuscripts Under Review

R1. **K. Younes**, H. Li, A. Majumdar, Y. Sun, T. Yoon, P. Ho, S. Wada, S. Song, T. Osaka, I. Inoue, D. Zhu, M. Dunne, D. Reis, L. Young, M. Ihme, "Acoustic fingerprints of radiation spurs in liquid water radiolysis," *Nature Physics*, 2026.

Refereed Conference Proceedings

C2. K. Ou, V. M. Perez-Ramirez, S. Cao, T. Yoon, H. Rajesh, D. Chakraborty, D. Singh, C. Redshaw, P. Dedeler, A. Oudin, E. Kur, **K. Younes**, M. Ihme, L. Lancia, C. Riconda, P. Michel, M. R. Edwards, "Optimization of high-damage-threshold transient gas optics for high-power lasers," *Optica Nonlinear Optics (NLO) 2025*, Technical Digest Series, paper Tu1B.3.

C1. **K. Younes**, A. Grenke, J.-P. Hickey, M. Gagnon, B. Elzein, "Enhanced Delayed Detached Eddy Simulations of Shock-Vector Control," *23rd AIAA International Space Planes and Hypersonic Systems and Technologies Conference*, AIAA 2020-2411. doi:10.2514/6.2020-2411

Theses

K. Younes, "Characterizing the picosecond dynamics of liquid water subject to hard X-ray irradiation," PhD thesis, Stanford University, 2026. purl:rw199cg3491

K. Younes, "Velocity scaling of high-speed turbulent boundary layer flows with wall heat transfer," Master's thesis, University of Waterloo, 2021. hdl:10012/17147

Conference Presentations and Talks

P14. A. Majumdar, H. Li, **K. Younes**, P. Muhunthan, G. Vignat, Y. Sun, S. Song, T. Sato, D. Zhu, M. Ihme, "Capturing nanometer-picosecond collective dynamics in supercritical water using split-pulse XPCS," *APS Global Physics Summit*, Anaheim, CA, Mar. 2025.

P13. T. Yoon, K. Ou, **K. Younes**, S. Cao, D. Singh, H. Rajesh, M. R. Edwards, M. Ihme, "Bayesian Optimization of Photochemically Induced Acousto-Optics in Ozone Gas for Enhanced Diffraction Efficiency," *APS Global Physics Summit*, Anaheim, CA, Mar. 2025.

P12. **K. Younes**, M. Poli, P. Muhunthan, S. Ermon, M. Ihme, "Rapid, Online Screening of Complex Phase Spaces Using Bayesian Optimization with Application to SAXS Measurements," *15th International Conference on Synchrotron Radiation Instrumentation*, Hamburg, Germany, Aug. 2024. **(Invited)**

P11. **K. Younes**, H. Li, Y. Sun, S. Song, T. Osaka, I. Inoue, M.-F. Lin, P. Ho, D. Zhu, M. Ihme, "Observing the Ultrafast Structural Response of Liquid Water Subject to Strong XFEL Radiation," *15th International Conference on Synchrotron Radiation Instrumentation*, Hamburg, Germany, Aug. 2024.

P10. **K. Younes**, H. Li, Y. Sun, S. Song, T. Osaka, I. Inoue, D. Zhu, M. Ihme, "Ultrafast Structural Reorganization of Supercritical Water Molecules Under XFEL Radiation," *APS Division of Atomic, Molecular and Optical Physics*, Fort Worth, TX, Jun. 2024.

P9. **K. Younes**, H. Li, Y. Sun, S. Song, T. Osaka, I. Inoue, D. Zhu, P. Ho, M. Ihme, "Probing ultrafast structural rearrangement in liquid water by X-ray experiments," *Thermofluids, Energy, and Propulsion Systems Seminar*, Stanford, CA, May 2024.

P8. **K. Younes**, M. Poli, P. Muhunthan, S. Ermon, M. Ihme, "Probing Supercritical Fluids: Rapid Exploration of Thermodynamic State Spaces," *Thermofluids, Energy, and Propulsion Systems Seminar*, Stanford, CA, Nov. 2023.

P7. **K. Younes**, M. Poli, P. Muhunthan, S. Ermon, M. Ihme, "Rapid, online screening of complex phase spaces using Bayesian Optimization for SAXS measurements," *Ultrafast Imaging and Tracking Instrumentation, Methods and Applications Conference*, Menlo Park, CA, Mar. 2023. **(Early Career Award)**

P6. **K. Younes** and J.-P. Hickey, "Mean-velocity scaling of compressible turbulent boundary layer flows under non-adiabatic wall conditions," *APS Division of Fluid Dynamics Meeting Abstracts*, Virtual, Nov. 2020.

P5. J.-P. Hickey, H. Daryan, **K. Younes**, "Dynamics of compound, compressible flow contractions," *APS Division of Fluid Dynamics Meeting Abstracts*, Virtual, Nov. 2020.

P4. M. Yao, **K. Younes**, D. Fan, J. Mouallem, J.-P. Hickey, "Targeted modal turbulent flow control via localized heating," *APS Division of Fluid Dynamics Meeting Abstracts*, Seattle, WA, Nov. 2019.

P3. **K. Younes** and J.-P. Hickey, "Effects of shear layer growth on the indirect noise in compound nozzles," *Thousand Islands Fluid Dynamics Meeting*, Gananoque, ON, Apr. 2019.

P2. **K. Younes**, M. Yao, J. Wang, J.-P. Hickey, "Towards predictive simulations for rocket propulsion," *Ontario Aerospace Council*, Toronto, ON, Mar. 2019.

P1. **K. Younes** and J.-P. Hickey, "Indirect noise in compound-compressible nozzle flows," *Thousand Islands Fluid Dynamics Meeting*, Gananoque, ON, Apr. 2018.

Academic Service

Mentoring

Liu Deng (PhD)	Fall 2025
Kshitij Ghormode (Master's)	Fall 2023, Winter 2024
Kai Rayle (Undergraduate)	Summer 2023
Aaron Grenke (Undergraduate)	Fall 2019

Reviewer

Progress in Energy and Combustion Science, *Journal of Fluid Mechanics*, *Physics of Fluids*, *Shock Waves*, *Journal of Aerospace Engineering*, *AIAA Journal*, *The Aeronautical Journal*, *IEEE Transactions on Aerospace and Electronic Systems*

Outreach

Instructor, seeME	2024
Demonstrator, STEMfest	2024
Student Ambassador, Waterloo Engineering	2015